



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CampusLoop : AI Intelligent Student Marketplace & Support System

The domain of the Project:
Genertive AI

Team Mentors (and their designation):

Sir. Prujith Radhkrishnan(Gen AI Mentor- Sure Trust)

Team Members:

Ms. GAYATRI SANJEEV MAHAJAN

Period of the project

May 2026 to July 2026



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled “CampusLoop” has been mentored by Sir.Prujith Radhkrishnan, organised by SURE Trust, from January 2026 to July 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. Gayatri Sanjeev Mahajan

Prujith Radhkrishnan
Gen AI Mentor— SURE

Prof. Radhakumari
Executive Director & Founder
SURE Trust



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Table of contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
7. Future Scope & Conclusion



Executive Summary

1. Project Background & Objectives

CampusLoop AI is a Generative AI-powered campus marketplace designed to provide a secure and intelligent platform for students to **buy, sell, rent, and exchange products** within their college community. Traditional online marketplaces often face problems such as fake product listings, unfair pricing, time-consuming negotiations, and lack of trust between buyers and sellers.

The main objective of this project is to solve these challenges by integrating **Generative AI** into the marketplace. AI is used to assist students with product listing, image verification, price estimation, intelligent negotiation, and chatbot support, making campus transactions safer, faster, and more reliable. Along with the marketplace, the platform also provides additional student services such as room/PG finder, teammate finder, notes sharing, turf booking, and other campus utilities.

2. Methodology & Technologies Used

The project is developed using a full-stack web architecture with **React.js (Vite)** for the frontend and **FastAPI (Python)** for the backend. **SQLite** with **SQLAlchemy** is used for database management.

The intelligent features are powered by **Google Gemini** and **Groq Llama** models. These models are integrated to provide AI-based image recognition, product price suggestions, negotiation assistance, chatbot responses, and content moderation. The platform also supports secure authentication, real-time chat, and efficient API communication to ensure a smooth user experience.

3. Key Outcomes

The developed system successfully provides a secure campus marketplace where students can list products for sale, rent, barter, or purchase with confidence. AI automatically assists users by verifying uploaded product images, suggesting appropriate selling prices, helping buyers and sellers during negotiations, and answering user queries through an intelligent chatbot.

These AI-powered features reduce fraudulent listings, improve pricing accuracy, simplify negotiations, and create a more trustworthy marketplace. The additional



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

campus utility modules further enhance the overall student experience by bringing multiple services into a single platform.

4. Recommendations & Future Scope

The platform can be further enhanced by integrating online payment gateways, multilingual support, voice-based AI interactions, mobile application support, and cloud-based database solutions such as PostgreSQL for better scalability. Future versions can also include advanced AI features such as personalized recommendations, semantic search using vector databases, and Retrieval-Augmented Generation (RAG) to provide more accurate and context-aware responses.

Project Highlights

CampusLoop AI is a Generative AI-powered campus marketplace that enhances the buying, selling, renting, and bartering experience for students by integrating intelligent features such as AI-based product price estimation, image verification, negotiation assistance, and chatbot support. The platform improves security, reduces fraudulent listings, simplifies student-to-student transactions, and provides a centralized solution for campus-related services. The project demonstrates how Generative AI can be effectively integrated into a real-world web application to create a secure, intelligent, and user-friendly digital marketplace for educational institutions.



Introduction

Background and Context

With the increasing use of online marketplaces, students frequently buy, sell, rent, or exchange items such as textbooks, calculators, electronics, lab equipment, and other academic resources. However, existing platforms are designed for the general public and often lack trust, student verification, and intelligent assistance. As a result, students face challenges such as fake listings, unfair pricing, lengthy negotiations, and difficulty finding reliable buyers or sellers.

CampusLoop AI is developed to solve these problems by providing a secure, college-exclusive marketplace where only verified students can participate. The platform integrates Generative AI to make transactions safer, faster, and more efficient. In addition to the marketplace, the platform also includes campus utility features such as room/PG finder, teammate finder, notes sharing, turf booking, and other student services.

Problem Statement and Project Goals

Students often spend a significant amount of time negotiating product prices, creating listings, and verifying whether a product is genuine. They may also become victims of fake product images, misleading descriptions, or fraudulent sellers.

The primary goal of CampusLoop AI is to develop an intelligent marketplace that uses Generative AI to improve the overall buying and selling experience. The project aims to:

- Provide a secure platform for buying, selling, renting, and bartering products.
- Automatically suggest fair product prices using AI.
- Verify uploaded product images to detect fake or stock images.
- Assist buyers and sellers during price negotiations through an AI negotiation assistant.
- Provide an AI chatbot to answer user queries and improve user interaction.
- Create a trusted and user-friendly campus marketplace with additional student utility services.



Scope and Limitations

Scope

The project provides a complete web-based marketplace where verified students can buy, sell, rent, or exchange products. It includes AI-powered features such as image verification, product price estimation, intelligent negotiation, chatbot assistance, and real-time chat between buyers and sellers. The platform also offers additional campus utility modules including room/PG finder, teammate finder, notes sharing, turf booking, and other community services.

Limitations

- The current system uses **SQLite**, which is suitable for development and small-scale deployment but is not ideal for large numbers of concurrent users.
- AI features depend on external APIs such as **Google Gemini** and **Groq**, requiring an internet connection.
- Online payment gateway integration has not been implemented in the current version.
- The platform is currently designed for use within a single educational institution using verified college email authentication.

Innovation Component

The main innovation of CampusLoop AI is the integration of **Generative AI** into a student marketplace to make online campus transactions more intelligent, secure, and user-friendly.

Unlike traditional marketplace applications, CampusLoop AI uses AI to automatically estimate product prices, verify uploaded images, assist in buyer-seller negotiations, and provide chatbot support. These intelligent features reduce manual effort, improve trust between users, and help prevent fraudulent activities.

By combining a secure campus marketplace with multiple AI-powered services on a single platform, CampusLoop AI demonstrates the practical application of Generative AI in solving real-world problems within educational institutions.



Project Objectives

Objectives and Goals

The main objectives of **CampusLoop AI** are:

- To develop a secure AI-powered campus marketplace where verified students can buy, sell, rent, and exchange products.
- To simplify the product listing process by using AI for image verification and automatic product price suggestions.
- To assist buyers and sellers during negotiations using an AI-powered negotiation assistant that helps both users reach a fair price.
- To provide an AI chatbot that answers user queries and improves the overall user experience.
- To reduce fraudulent listings by detecting fake or stock images using AI-based image analysis.
- To enable secure real-time communication between buyers and sellers through an integrated chat system.
- To provide additional campus utility services such as room/PG finder, teammate finder, notes sharing, turf booking, and other student support features on a single platform.

Expected Outcomes and Deliverables

The expected outcomes and deliverables of the project include:

- A fully functional AI-powered campus marketplace web application.
- A **FastAPI (Python)** backend with REST APIs and real-time chat support.
- A **React.js (Vite)** frontend providing a responsive and user-friendly interface.
- AI-powered features including product price estimation, image verification, chatbot support, and negotiation assistance.
- Secure user authentication and product management for buying, selling, renting, and bartering.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

- Real-time communication between buyers and sellers through the chat module.
- A centralized platform that combines marketplace services with additional campus utility features to improve student convenience and collaboration.



Methods and Technology Used

The project follows a full-stack web development approach integrated with Generative AI technologies.

- **Frontend:** React.js (Vite) is used to develop a responsive and interactive user interface.
- **Backend:** FastAPI (Python) is used to build REST APIs and manage the business logic.
- **Database:** SQLite with SQLAlchemy ORM is used to store user details, product listings, chat messages, and transaction data.
- **Real-Time Communication:** WebSockets are used to enable live chat between buyers and sellers.
- **Generative AI:** Google Gemini and Groq Llama models are integrated to provide intelligent features such as image verification, product price estimation, AI-assisted negotiation, chatbot responses, and content moderation.
- **Authentication:** Secure user authentication is implemented to ensure that only verified users can access the platform.

Tools and Software Used

- Visual Studio Code (IDE)
- React.js (Vite)
- FastAPI (Python)
- SQLite
- SQLAlchemy
- Google Gemini API
- Groq API (Llama Models)
- Postman



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

- Git & GitHub
- Node.js and NPM
- Python Virtual Environment (venv)

Data Collection Approach

The application uses data generated by users during normal platform usage. Users create accounts, upload product listings, exchange messages, and perform buying, selling, renting, and bartering activities. Product images uploaded by users are analyzed using AI for authenticity checks, while product details are processed to generate price suggestions and support AI-assisted negotiation.

Project Architecture

The architecture of CampusLoop AI consists of four main layers:

1. Frontend Layer

The frontend is developed using React.js and provides interfaces for user authentication, marketplace browsing, product listing, chat, negotiation, and profile management. User requests are sent to the backend through REST APIs and WebSockets.

2. Backend Layer

The FastAPI backend manages authentication, user accounts, product listings, chat functionality, AI service integration, and transaction processing. It acts as the central controller of the application.

3. AI Layer

The AI layer integrates Google Gemini and Groq Llama models to provide intelligent features including:

- AI product price suggestion
- AI image verification
- AI-assisted negotiation
- AI chatbot
- Chat content moderation



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

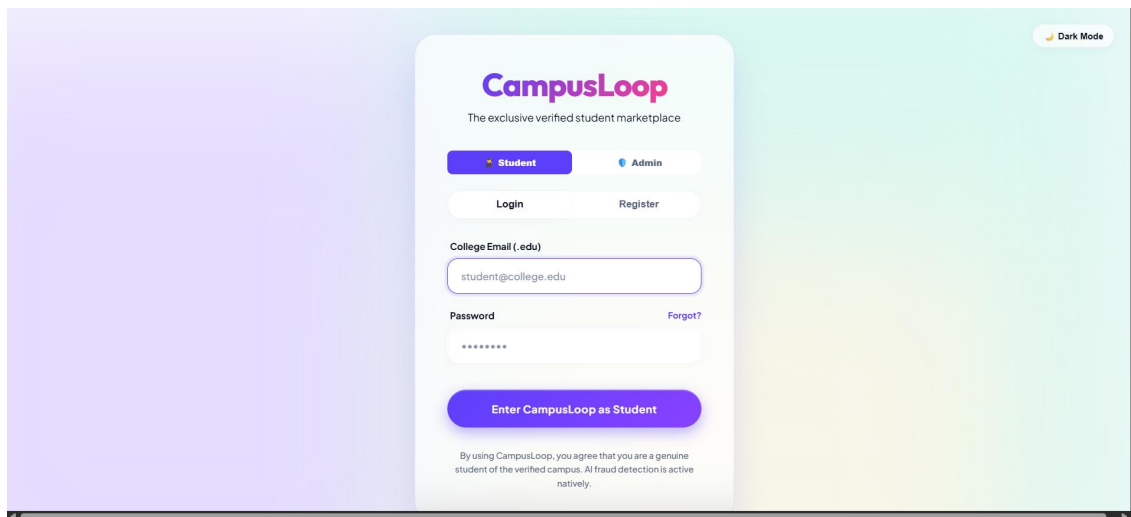
The backend sends the required data to the AI models, processes their responses, and returns the results to the user interface.

4. Database Layer

SQLite stores all application data including user information, product listings, chat messages, negotiations, and transaction history. SQLAlchemy is used for database operations.

Final Project Working Screenshots with Explanation

Screenshot 1 – Home Page

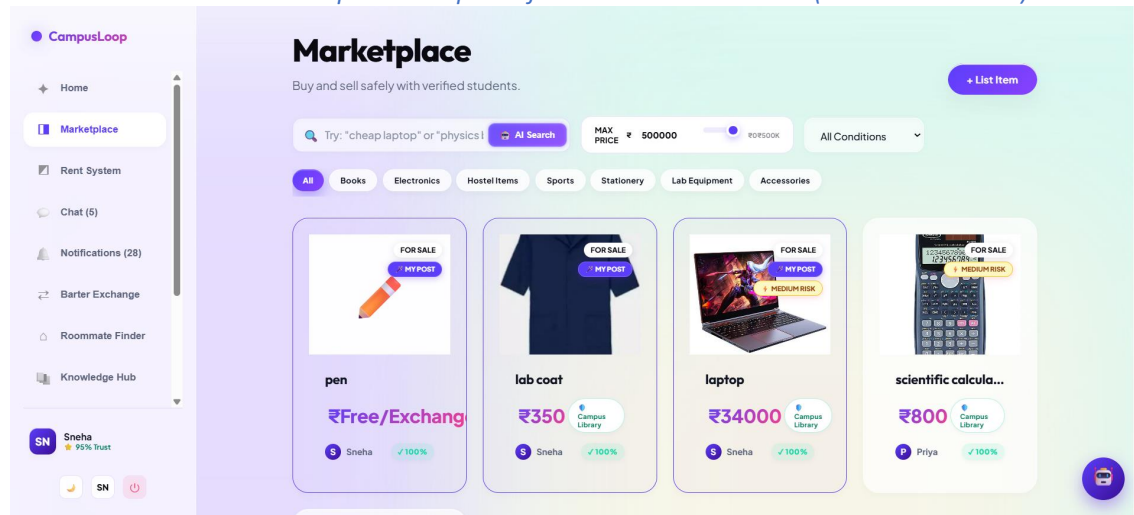


Explanation: Displays the landing page where users can log in, register, and explore the CampusLoop AI platform.

Screenshot 2 – Marketplace

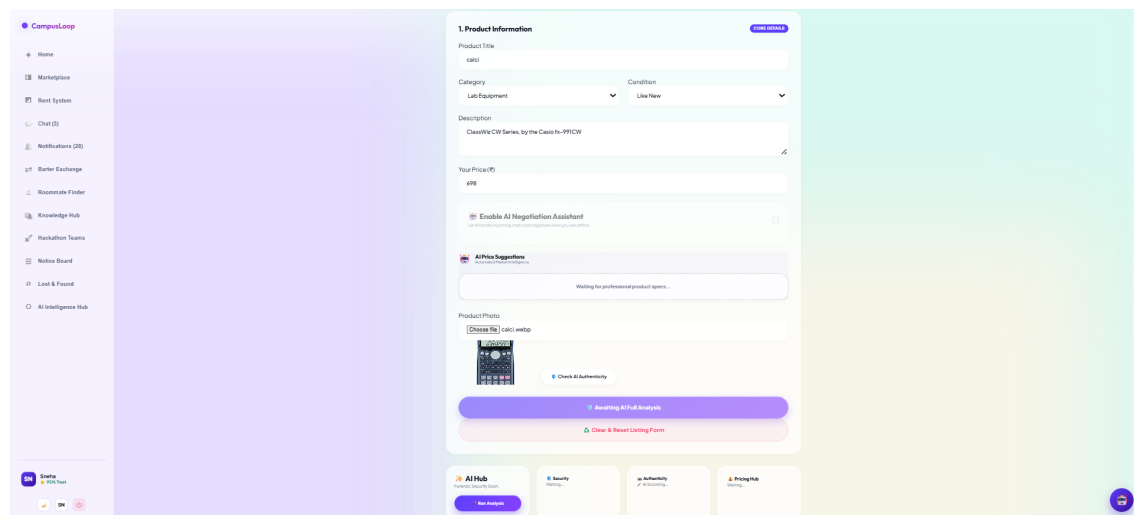


Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)



Explanation: Shows all available products where students can buy, sell, rent, or exchange items securely.

Screenshot 3 – Product Listing with AI Price Suggestion



Explanation: Users upload product details and images. AI analyzes the information and suggests an appropriate selling price.

Screenshot 4 – AI Image Verification

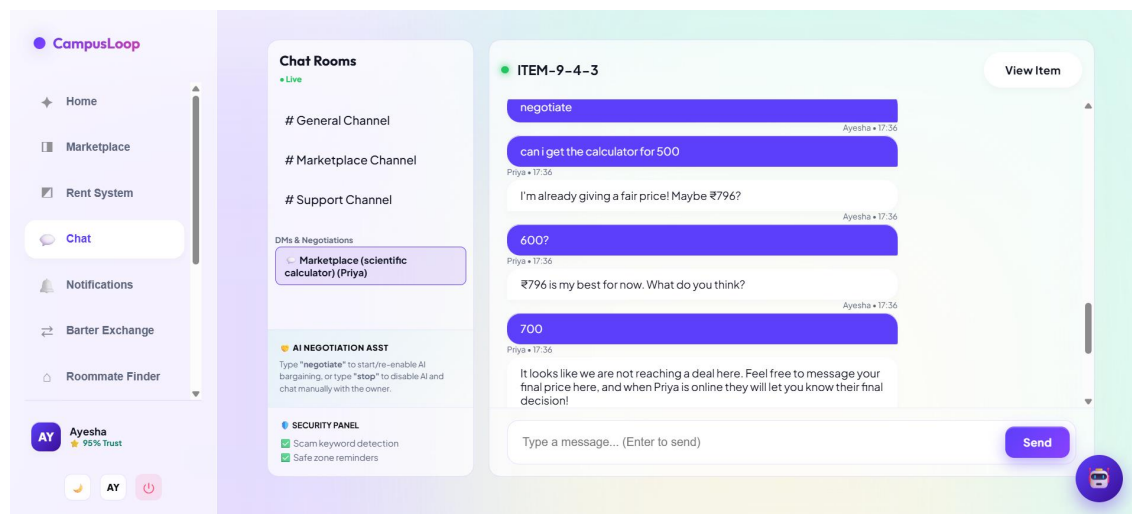


Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)



Explanation: After uploading a product image, AI checks whether the image appears genuine or resembles a stock or fake image, helping reduce fraudulent listings.

Screenshot 5 – AI Negotiation Chat

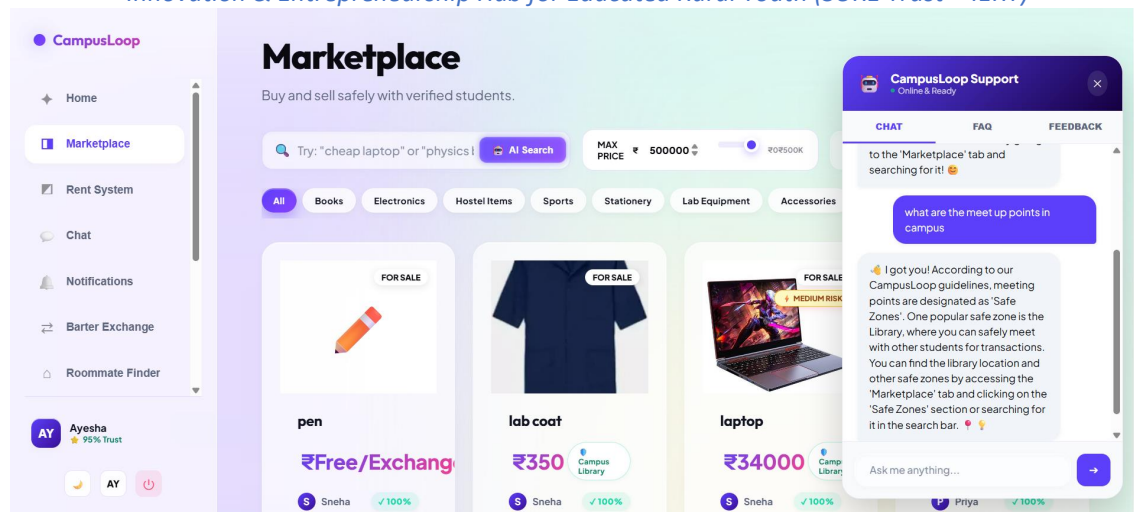


Explanation: Buyers and sellers communicate through the chat system. The AI negotiation assistant suggests fair counter-offers and helps both users reach a mutually acceptable price.

Screenshot 6 – AI Chatbot

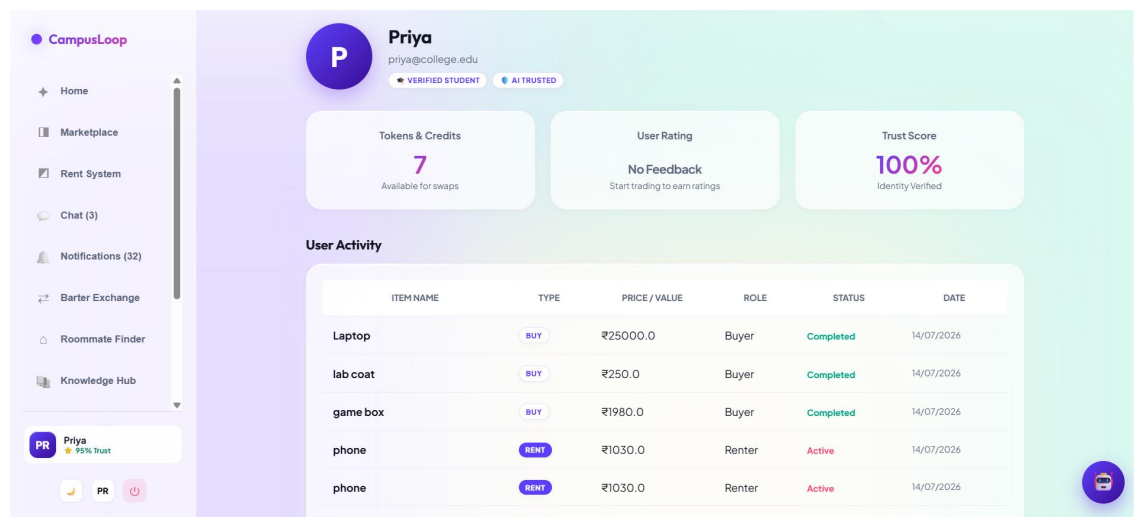


Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)



Explanation: The chatbot answers user queries, provides guidance about platform features, and assists users during marketplace interactions.

Screenshot 7 – User Profile & Transaction History



Explanation: Displays user profile information, listed products, completed transactions, and other account details.

Project GitHub Link

GitHub Repository:

<https://github.com/sure-trust/GAYATRI-SANJEEV-MAHAJAN-g4-gen-ai>



New Learnings

During the development of **CampusLoop AI**, I gained practical knowledge of full-stack web development and Generative AI integration. I learned how to build a complete web application using **React.js** for the frontend and **FastAPI** for the backend. I also learned how to manage data using **SQLite** and **SQLAlchemy**.

One of the most valuable learning experiences was integrating **Google Gemini** and **Groq Llama** APIs to implement AI-powered features such as product price suggestion, image verification, AI-assisted negotiation, chatbot support, and content moderation. I also learned how to use **WebSockets** to enable real-time communication between buyers and sellers.

Throughout the project, I improved my understanding of API integration, debugging, database management, authentication, GitHub version control, and testing using Postman. In addition to technical skills, I developed better project planning, problem-solving, documentation, and time management skills while completing different modules of the project.

Overall Experience

Developing **CampusLoop AI** was a valuable and rewarding learning experience. It gave me the opportunity to apply both web development and Generative AI concepts to solve real-world problems faced by students during campus transactions.

While developing the project, I encountered several challenges, including integrating AI APIs, handling backend and frontend communication, implementing real-time chat, and debugging application errors. Solving these challenges helped improve my technical knowledge and confidence in developing AI-powered applications.

Overall, this project strengthened my skills in software development, Generative AI integration, problem-solving, and project management. It also provided practical experience in building a complete full-stack application from planning and development to testing and documentation.



Conclusion and Future Scope

Conclusion

The main objective of **CampusLoop AI** was to develop a secure and intelligent campus marketplace that simplifies buying, selling, renting, and exchanging products among students. The project successfully integrates **Generative AI** to enhance the overall marketplace experience through intelligent features such as AI-powered product price suggestion, image verification, chatbot assistance, AI-supported negotiation, and real-time communication between buyers and sellers.

The platform provides a user-friendly interface with secure authentication, efficient product management, and real-time interactions, making campus transactions more convenient and trustworthy. In addition to the marketplace, the application also offers student utility features such as room/PG finder, teammate finder, notes sharing, and turf booking, creating a centralized platform for various campus activities.

Overall, the project demonstrates how Generative AI can be effectively integrated into a real-world web application to improve security, reduce manual effort, and enhance user experience.

Future Scope

The project can be further enhanced by implementing the following features:

- Integration of secure online payment gateways such as UPI or Razorpay for seamless digital transactions.
- Deployment of the application on cloud platforms with PostgreSQL or MySQL for better scalability and multi-user support.
- Development of Android and iOS mobile applications to improve accessibility.
- Addition of multilingual and voice-based AI support to make the platform more user-friendly.
- Personalized product recommendations based on user preferences and browsing history using AI.
- More advanced AI models for image analysis, negotiation, and chatbot responses to improve accuracy and user experience.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

- Expansion of the platform to support multiple colleges and universities through institution-specific authentication and administration.

The future enhancements will make CampusLoop AI more scalable, intelligent, and suitable for large-scale deployment across educational institutions.